

Example 2.2.

If $L[f(t)] = F(S)$, Then $L^{-1}[F(S)] = f(t)$

Where L^{-1} is called the inverse Laplace transform operator.

$$(1) L^{-1}\left(\frac{1}{S}\right) = 1 \left[\because (1) L[f(t)] = L(1) = \frac{1}{S} \right]$$

$$(2) L^{-1}\left(\frac{a}{S}\right) = a \left[\because L(f(t)) = L(a) = \frac{a}{S} \right]$$

$$(3) L^{-1}\left(\frac{1}{S^2}\right) = t \left[\because L(f(t)) = L(t) = \frac{1}{S^2} \right]$$

$$(4) L^{-1}\left(\frac{1}{S-a}\right) = e^{at} \left[\because L(f(t)) = L(e^{at}) = \frac{1}{S-a} \right]$$

$$(5) L^{-1}\left(\frac{1}{S-2}\right) = e^{2t} \left[\because L(f(t)) = L(e^{2t}) = \frac{1}{S-2} \right]$$

$$(6) L^{-1}\left(\frac{a}{S^2 + a^2}\right) = \sin at \left[\because L(f(t)) = L(\sin at) = \frac{a}{S^2 + a^2} \right]$$

$$(7) L^{-1}\left(\frac{2}{S^2 + 2^2}\right) = \sin 2t \left[\because L(f(t)) = L(\sin 2t) = \frac{2}{S^2 + 2^2} \right]$$

$$(8) L^{-1}\left(\frac{S}{S^2 + a^2}\right) = \cos at \left[\because L(f(t)) = L(\cos at) = \frac{S}{S^2 + a^2} \right]$$

$$(9) L^{-1}\left(\frac{S}{S^2 + 2^2}\right) = \cos 2t \left[\because L(f(t)) = L(\cos 2t) = \frac{S}{S^2 + 2^2} \right]$$

$$(10) L^{-1}\left(\frac{a}{S^2 - a^2}\right) = \sin hat \left[\because L(f(t)) = L(\sin hat) = \frac{a}{S^2 - a^2} \right]$$

$$(11) L^{-1}\left(\frac{2}{S^2 - 2^2}\right) = \sin h2t \left[\because L(f(t)) = L(\sin h2t) = \frac{2}{S^2 - 2^2} \right]$$

$$(12) L^{-1}\left(\frac{S}{S^2 - a^2}\right) = \cos hat \left[\because L(f(t)) = L(\cos hat) = \frac{S}{S^2 - a^2} \right]$$

$$(13) L^{-1}\left(\frac{S}{S^2 - 2^2}\right) = \cos h2t \left[\because L(f(t)) = L(\cos h2t) = \frac{S}{S^2 - 2^2} \right]$$

Example 58: Find Laplace Transform of $f(t) = e^{at}$

Answer:

We have,

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt$$

Given,

$$f(t) = e^{at}$$

$$\begin{aligned} \therefore L(e^{at}) &= \int_0^{\infty} e^{at} \cdot e^{-st} dt \\ &= \int_0^{\infty} e^{at-st} dt \end{aligned}$$

$$\begin{aligned} &= \int_0^{\infty} e^{-st+at} dt \\ &= \int_0^{\infty} e^{-(st-at)} dt \\ &= \int_0^{\infty} e^{-(s-a)t} dt \\ &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} \\ &= \frac{1}{-(s-a)} \left[e^{-\infty} - e^{-0} \right] \\ &= -\frac{1}{(s-a)} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right] \\ &= -\frac{1}{(s-a)} \left[\frac{1}{\infty} - \frac{1}{1} \right] \\ &= -\frac{1}{(s-a)} [0 - 1] \\ &= \frac{1}{s-a} \text{ Answer} \end{aligned}$$

$$\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a} \text{ Answer}$$

$$\left[\int e^{-mx} dx = \frac{e^{-mx}}{-m} \right]$$

$$[e^{\infty} = \infty]$$

$$\left[\frac{1}{\infty} = 0 \right]$$

Example 64: Find $L\{e^{-at}t\} = ?$

Answer:

Here, $f(t) = t$

The first shift theorem states that,

If $L\{f(t)\} = f(s)$ -----(i)

Then $L\{e^{-at}f(t)\} = f(s + a)$ -----(ii)

We have, according to equation no (i), $L(f(t)) = L(t) = \frac{1}{s^2}$ [Here $f(s) = \frac{1}{s^2}$]

If $f(s) = \frac{1}{s^2}$

$\therefore f(s + a) = \frac{1}{(s + a)^2}$

Hence, according to equation no (ii), we can write

$L\{e^{-at}t\} = f(s + a) = \frac{1}{(s + a)^2}$ Answer

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Example 73: Find Laplace Transform of $t^2 \sin 2t$

Here, $f(t) = \sin 2t$

This theorem states that

If $L\{f(t)\} = f(s)$ -----(i)

Then, $L\{t^2 \cdot f(t)\} = (-1)^2 \frac{d^2}{ds^2} \{f(s)\}$ -----(ii)

We have, according to equation no (i), $L(f(t)) = L(\sin 2t) = \frac{2}{s^2 + 2^2} = \frac{2}{s^2 + 4}$

[Here $f(s) = \frac{2}{s^2 + 4}$]

Hence, according to equation no (ii), we can write

$L\{t^2 \cdot f(t)\} = (-1)^2 \frac{d^2}{ds^2} \{f(s)\}$

$$L\{t^2 \cdot \sin 2t\} = (-1)^2 \frac{d^2}{ds^2} (f(s)) \quad [f(t) = \sin 2t]$$

$$L\{t^2 \cdot \sin 2t\} = (-1)^2 \frac{d^2}{ds^2} \left\{ \frac{2}{s^2 + 4} \right\} \quad [f(s) = \frac{2}{s^2 + 4}]$$

$$L\{t^2 \cdot \sin 2t\} = + \frac{d}{ds} \left[\frac{d}{ds} \left\{ \frac{2}{s^2 + 4} \right\} \right]$$

$$L\{t^2 \cdot \sin 2t\} = \frac{d}{ds} \left[\frac{d}{ds} \{ 2(s^2 + 4)^{-1} \} \right]$$

$$L\{t^2 \cdot \sin 2t\} = 2 \frac{d}{ds} \left[(-1)(s^2 + 4)^{-1-1} \cdot \frac{d}{ds} (s^2 + 4) \right] \quad \left[\frac{d}{dx} (x^n) = nx^{n-1} \right]$$

$$L\{t^2 \cdot \sin 2t\} = 2 \frac{d}{ds} \left[(-1)(s^2 + 4)^{-2} \cdot (2s) \right]$$

$$L\{t^2 \cdot \sin 2t\} = -4 \frac{d}{ds} \left[(s^2 + 4)^{-2} \cdot s \right]$$

$$L\{t^2 \cdot \sin 2t\} = -4 \left[s \frac{d}{ds} \left[(s^2 + 4)^{-2} \right] + \left[(s^2 + 4)^{-2} \right] \frac{d}{ds} (s) \right]$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \right\} v dx dx]$$

$$L\{t^2 \cdot \sin 2t\} = -4 \left[s \left[(-2)(s^2 + 4)^{-2-1} \cdot \frac{d}{ds} (s^2 + 4) \right] + \left[(s^2 + 4)^{-2} \right] \cdot 1 \right]$$

$$L\{t^2 \cdot \sin 2t\} = -4 \left[s \left[(-2)(s^2 + 4)^{-3} \cdot (2s) \right] + \left[(s^2 + 4)^{-2} \right] \right]$$

$$L\{t^2 \cdot \sin 2t\} = -4 \left[s \left[\frac{-4s}{(s^2 + 4)^3} \right] + \left[\frac{1}{(s^2 + 4)^2} \right] \right]$$

$$L\{t^2 \cdot \sin 2t\} = \left[\left[\frac{16s^2}{(s^2 + 4)^3} \right] - \left[\frac{4}{(s^2 + 4)^2} \right] \right]$$

$$L\{t^2 \cdot \sin 2t\} = \frac{16s^2}{(s^2 + 4)^3} - \frac{4}{(s^2 + 4)^2}$$

$$L\{t^2 \cdot \sin 2t\} = \frac{16s^2 - 4(s^2 + 4)}{(s^2 + 4)^3}$$

$$L\{t^2 \cdot \sin 2t\} = \frac{16s^2 - 4s^2 - 16}{(s^2 + 4)^3}$$

$$L\{t^2 \cdot \sin 2t\} = \frac{12s^2 - 16}{(s^2 + 4)^3} \text{ Answer}$$

Example 74: Find the Laplace Transform $L\{t^2 \cos 3t\}$

Answer:

Here, $f(t) = \cos 3t$

This theorem states that,

$$\text{If } L\{f(t)\} = f(s) \quad \text{----- (i)}$$

$$\text{Then } L\{t^2 f(t)\} = (-1)^2 \frac{d^2}{ds^2} \{f(s)\} \quad \text{----- (ii)}$$

$$\text{We have, according to equation no (i), } L\{f(t)\} = L(\cos 3t) = \frac{s}{s^2 + 3^2} = \frac{s}{s^2 + 9}$$

$$[\text{Here } f(s) = \frac{s}{s^2 + a^2}]$$

Hence, according to equation no (ii), we can write,

$$L\{t^2 \cdot f(t)\} = (-1)^2 \frac{d^2}{ds^2} \{f(s)\}$$

$$L\{t^2 \cdot \cos 3t\} = (-1)^2 \frac{d^2}{ds^2} \{f(s)\} \quad [f(t) = \cos 3t]$$

$$L\{t^2 \cdot \cos 3t\} = (-1)^2 \frac{d^2}{ds^2} \left\{ \frac{s}{s^2 + 9} \right\}$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[\frac{d}{ds} \left\{ \frac{s}{s^2 + 9} \right\} \right]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[\frac{d}{ds} \{s(s^2 + 9)^{-1}\} \right]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[s \frac{d}{ds} \{(s^2 + 9)^{-1}\} + (s^2 + 9)^{-1} \frac{d}{ds} s \right]$$

$$[\because \int uv dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \right\} v dx dx]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[s \{(-1)(s^2 + 9)^{-1-1}\} \cdot \frac{d}{ds} (s^2 + 9) + (s^2 + 9)^{-1} \cdot 1 \right]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[-s(s^2 + 9)^{-2} \cdot (2s) + (s^2 + 9)^{-1} \right]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[-2s^2 (s^2 + 9)^{-2} + (s^2 + 9)^{-1} \right]$$

$$L\{t^2 \cdot \cos 3t\} = \frac{d}{ds} \left[-2s^2 (s^2 + 9)^{-2} \right] + \frac{d}{ds} \left[(s^2 + 9)^{-1} \right]$$

$$L\{t^2 \cdot \cos 3t\} = -2 \frac{d}{ds} \left[s^2 (s^2 + 9)^{-2} \right] + \frac{d}{ds} \left[(s^2 + 9)^{-1} \right]$$

$$L\{t^2 \cdot \cos 3t\} = -2 \left[s^2 \frac{d}{ds} \left[(s^2 + 9)^{-2} \right] + (s^2 + 9)^{-2} \frac{d}{ds} s^2 \right] + \frac{d}{ds} \left[(s^2 + 9)^{-1} \right]$$

$$L\{t^2 \cdot \cos 3t\} = -2 \left[s^2 \cdot (-2) \cdot (s^2 + 9)^{-2-1} \frac{d}{ds} (s^2 + 9) + (s^2 + 9)^{-2} \cdot (2s) \right] + (-1)(s^2 + 9)^{-1-1} \frac{d}{ds} [(s^2 + 9)]$$

$$L\{t^2 \cdot \cos 3t\} = -2 \left[s^2 \cdot (-2) \cdot (s^2 + 9)^{-3} \cdot (2s) + (s^2 + 9)^{-2} \cdot (2s) \right] + (-1)(s^2 + 9)^{-2} \cdot (2s)$$

$$L\{t^2 \cdot \cos 3t\} = -2 \left[-4s^3 (s^2 + 9)^{-3} + 2s(s^2 + 9)^{-2} \right] - 2s(s^2 + 9)^{-2}$$

$$L\{t^2 \cdot \cos 3t\} = 8s^3 (s^2 + 9)^{-3} - 4s(s^2 + 9)^{-2} - 2s(s^2 + 9)^{-2}$$

$$L\{t^2 \cdot \cos 3t\} = 8s^3 (s^2 + 9)^{-3} - 6s(s^2 + 9)^{-2}$$

$$L\{t^2 \cdot \cos 3t\} = \frac{8s^3}{(s^2 + 9)^3} - \frac{6s}{(s^2 + 9)^2}$$

$$L\{t^2 \cdot \cos 3t\} = \frac{8s^3 - 6s(s^2 + 9)}{(s^2 + 9)^3}$$

$$L\{t^2 \cdot \cos 3t\} = \frac{8s^3 - 6s^3 - 54s}{(s^2 + 9)^3}$$

$$L\{t^2 \cdot \cos 3t\} = \frac{2s^3 - 54s}{(s^2 + 9)^3} \quad \text{Answer}$$

Example 75: Find Laplace Transform of $\left\{\frac{\sin 2t}{t}\right\}$

Answer:

Here, $f(t) = \sin 2t$

Division theorem states that,

$$\text{If } \mathcal{L}\{f(t)\} = f(s) \quad \text{-----(i)}$$

$$\text{Then } \mathcal{L}\left[\frac{1}{t} f(t)\right] = \int_s^\infty f(s) ds \quad \text{-----(ii)}$$

We have, according to equation no (i), $\mathcal{L}(f(t)) = \mathcal{L}(\sin 2t) = \frac{2}{s^2 + 4}$ [Here

$$f(s) = \frac{2}{s^2 + 4}]$$

Hence, according to equation no (ii), we can write

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \int_s^\infty f(s) ds$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = \int_s^\infty f(s) ds$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = \int_s^\infty \frac{2}{s^2 + 4} ds$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = \int_s^\infty \frac{2}{s^2 + 2^2} ds$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \int_s^\infty \frac{1}{s^2 + 2^2} ds$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \times \frac{1}{2} \left[\tan^{-1} \frac{s}{2} \right]_s^\infty \quad \left[\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} \right]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \times \frac{1}{2} \left[\tan^{-1} \frac{\infty}{2} - \tan^{-1} \frac{s}{2} \right]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \times \frac{1}{2} \left[\tan^{-1} \infty - \tan^{-1} \frac{s}{2} \right]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \times \frac{1}{2} \left[\tan^{-1} \tan \frac{\pi}{2} - \tan^{-1} \frac{s}{2} \right] \quad [\because \tan \frac{\pi}{2} = \infty]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = 2 \times \frac{1}{2} \left[\frac{\pi}{2} - \tan^{-1} \frac{s}{2} \right]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = \left[\frac{\pi}{2} - \tan^{-1} \frac{s}{2} \right]$$

$$\mathcal{L}\left[\frac{1}{t} f(t)\right] = \mathcal{L}\left[\frac{1}{t} \sin 2t\right] = \left[\frac{\pi}{2} - \tan^{-1} \frac{s}{2} \right] \quad \text{Answer}$$

Inverese Laplaces transform

Example 100 Find the inverse Laplace transforms of the following:

(i) $\frac{1}{s-2}$, (ii) $\frac{s}{s^2-16}$, (iii) $\frac{5}{s^2+25}$, (iv) $\frac{1}{(s+3)^2-4}$, (v) $\frac{1}{2s-7}$

Answers:

(i) Given $f(s) = \frac{1}{s-2}$

We have, $\therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$ [Example 58]

$$\therefore e^{at} = L^{-1}\left(\frac{1}{s-a}\right)$$

$$\therefore e^{2t} = L^{-1}\left(\frac{1}{s-2}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \text{ Answer}$$

(ii) $L^{-1}\left(\frac{s}{s^2-16}\right) = ?$

We have, $\therefore L(f(t)) = L(\cosh at) = \frac{s}{s^2-a^2}$ [Example 63]

$$\therefore \cosh at = L^{-1}\left(\frac{s}{s^2-a^2}\right)$$

$$\therefore \cosh 4t = L^{-1}\left(\frac{s}{s^2-4^2}\right)$$

$$\therefore L^{-1}\left(\frac{s}{s^2-4^2}\right) = \cosh 4t \text{ Answer}$$

(iii) We have, $\therefore L(f(t)) = L(\sin at) = \frac{a}{s^2+a^2}$ [Example 60]

$$\therefore \sin at = L^{-1}\left(\frac{a}{s^2+a^2}\right)$$

$$\therefore \sin 5t = L^{-1}\left(\frac{5}{s^2+5^2}\right)$$

$$\therefore L^{-1}\left(\frac{5}{s^2+5^2}\right) = \sin 5t$$

$$\therefore L^{-1}\left(\frac{5}{s^2+5^2}\right) = \sin 5t \text{ Answer}$$

Fourier Transform

Summary:

01. The Fourier transform transforms a function of t (in the time domain) into a function of ω (in the frequency domain), and the inverse Fourier transformation does the reverse, $\mathbf{g}(\omega)$ is also called the spectrum function of $\mathbf{f}(t)$
02. We call $\mathbf{g}(\omega)$ the direct Fourier Transform of $\mathbf{f}(t)$ and $\mathbf{f}(t)$ the inverse Fourier Transform of $\mathbf{g}(\omega)$. The Transform $\mathbf{g}(\omega)$ is the frequency –domain specification of $\mathbf{f}(t)$

03. Fourier Transform: $\mathbf{g}(\omega) = \int_{-\infty}^{\infty} \mathbf{f}(t) e^{-i\omega t} dt$

04. Inverse Fourier Transform: $\mathbf{f}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [\mathbf{g}(\omega)] e^{i\omega t} d\omega$

Example 41: Find Fourier Transform of

$$\begin{aligned} f(t) &= 1 && ; 0 \leq t < 1 \\ &= -1 && ; -1 \leq t < 0 \\ &= 0 && ; |t| > 1 \end{aligned}$$

We have $g(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$

$$g(\omega) = \int_{-\infty}^{-1} f(t)e^{-i\omega t} dt + \int_{-1}^0 f(t)e^{-i\omega t} dt + \int_0^1 f(t)e^{-i\omega t} dt + \int_1^{\infty} f(t)e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^{-1} 0 \cdot e^{-i\omega t} dt + \int_{-1}^0 (-1)e^{-i\omega t} dt + \int_0^1 1 \cdot e^{-i\omega t} dt + \int_1^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$g(\omega) = -\int_{-1}^0 e^{-i\omega t} dt + \int_0^1 e^{-i\omega t} dt$$

$$g(\omega) = -\left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-1}^0 + \left[\frac{e^{-i\omega t}}{-i\omega} \right]_0^1 \quad \left[\because \int e^{-mx} dx = \frac{e^{-mx}}{-m} \right]$$

$$g(\omega) = -\left[\frac{e^{-i\omega \cdot 0}}{-i\omega} - \frac{e^{-i\omega(-1)}}{-i\omega} \right] + \left[\frac{e^{-i\omega \cdot 1}}{-i\omega} - \frac{e^{-i\omega \cdot 0}}{-i\omega} \right]$$

$$g(\omega) = -\left[\frac{e^{-0}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{e^{-0}}{-i\omega} \right]$$

$$g(\omega) = -\left[\frac{\frac{1}{e^0}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{\frac{1}{e^0}}{-i\omega} \right]$$

$$g(\omega) = -\left[\frac{\frac{1}{1}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{\frac{1}{1}}{-i\omega} \right]$$

$$g(\omega) = -\left[\frac{1}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right] + \left[\frac{e^{-i\omega}}{-i\omega} - \frac{1}{-i\omega} \right]$$

$$g(\omega) = \left[\frac{1}{i\omega} - \frac{e^{i\omega}}{i\omega} \right] + \left[-\frac{e^{-i\omega}}{i\omega} + \frac{1}{i\omega} \right]$$

$$g(\omega) = \frac{1}{i\omega} + \frac{1}{i\omega} - \frac{e^{i\omega}}{i\omega} - \frac{e^{-i\omega}}{i\omega}$$

$$g(\omega) = \frac{2}{i\omega} - \frac{1}{i\omega} (e^{i\omega} + e^{-i\omega})$$

$$g(\omega) = \frac{2}{i\omega} - \frac{1}{i\omega} \frac{2}{2} (e^{i\omega} + e^{-i\omega})$$

$$g(\omega) = \frac{2}{i\omega} - \frac{2}{i\omega} \frac{1}{2} (e^{i\omega} + e^{-i\omega})$$

$$g(\omega) = \frac{2}{i\omega} - \frac{2}{i\omega} \cos \omega \quad \left[\because \cos x = \frac{1}{2} (e^{ix} + e^{-ix}) \right]$$

$$g(\omega) = \frac{2}{i\omega} (1 - \cos \omega) \text{ Answer}$$

Example 42: Find Fourier Transform of $f(t) = e^{-|t|}$

Or

Find Fourier Transform of

$$f(t) = e^{-t} \quad ; t > 0$$

$$= e^t \quad ; t < 0$$

Answer:

Given

$$f(t) = e^{-t} \quad ; t > 0$$

$$= e^t \quad ; t < 0$$

----- (i)

We have,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^0 f(t) e^{-i\omega t} dt + \int_0^{\infty} f(t) e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^0 e^t e^{-i\omega t} dt + \int_0^{\infty} e^{-t} e^{-i\omega t} dt \quad [\text{Given equation no (i)}]$$

$$g(\omega) = \int_{-\infty}^0 e^t e^{-i\omega t} dt + \int_0^{\infty} e^{-t} e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^0 e^{t-i\omega t} dt + \int_0^{\infty} e^{-t-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^0 e^{(1-i\omega)t} dt + \int_0^{\infty} e^{-(1+i\omega)t} dt$$

$$g(\omega) = \left[\frac{e^{(1-i\omega)t}}{(1-i\omega)} \right]_{-\infty}^0 + \left[\frac{e^{-(1+i\omega)t}}{-(1+i\omega)} \right]_0^{\infty}$$

$$g(\omega) = \left[\frac{e^{(1-i\omega).0}}{(1-i\omega)} - \frac{e^{(1-i\omega)(-\infty)}}{(1-i\omega)} \right] + \left[\frac{e^{-(1+i\omega)\infty}}{-(1+i\omega)} - \frac{e^{-(1+i\omega).0}}{-(1+i\omega)} \right]$$

$$\begin{aligned}
g(\omega) &= \left[\frac{e^0}{(1-i\omega)} - \frac{e^{-\infty}}{(1-i\omega)} \right] + \left[\frac{e^{-\infty}}{-(1+i\omega)} - \frac{e^{-0}}{-(1+i\omega)} \right] \\
g(\omega) &= \left[\frac{1}{(1-i\omega)} - \frac{1}{e^{\infty}(1-i\omega)} \right] + \left[\frac{e^{-\infty}}{-(1+i\omega)} - \frac{e^{-0}}{-(1+i\omega)} \right] \\
g(\omega) &= \frac{1}{(1-i\omega)} \left[1 - \frac{1}{e^{\infty}} \right] + \frac{-1}{1+i\omega} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right] \\
g(\omega) &= \frac{1}{(1-i\omega)} \left[1 - \frac{1}{\infty} \right] + \frac{-1}{1+i\omega} \left[\frac{1}{\infty} - \frac{1}{1} \right] \\
g(\omega) &= \frac{1}{(1-i\omega)} [1-0] + \frac{-1}{1+i\omega} \left[\frac{1}{\infty} - \frac{1}{1} \right] \\
g(\omega) &= \frac{1}{(1-i\omega)} + \frac{-1}{1+i\omega} [0-1] \\
g(\omega) &= \frac{1}{(1-i\omega)} + \frac{1}{1+i\omega} \\
g(\omega) &= \frac{1+i\omega+1-i\omega}{(1-i\omega)(1+i\omega)} \\
g(\omega) &= \frac{2}{(1-i\omega)(1+i\omega)} \\
g(\omega) &= \frac{2}{(1-i^2\omega^2)} \\
g(\omega) &= \frac{2}{1+\omega^2} \quad [i^2 = -1]
\end{aligned}$$

Answer

Example 43: Find Fourier Transform for the given functions

$$\begin{aligned}
f(t) &= e^{-2t} & ; t \geq 0 \\
f(t) &= 0 & ; t < 0
\end{aligned}$$

Answer:

Given

$$\begin{aligned}
f(t) &= e^{-2t} & ; t \geq 0 \\
f(t) &= 0 & ; t < 0
\end{aligned}$$

We have,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$g(\omega) = \int_{-\infty}^0 f(t) e^{-i\omega t} dt + \int_0^{\infty} f(t) e^{-i\omega t} dt$$

------(i)

$$g(\omega) = \int_{-\infty}^0 0 \cdot e^{-i\omega t} dt + \int_0^{\infty} e^{-2t} e^{-i\omega t} dt \quad [\text{Given equation no (i)}]$$

$$g(\omega) = \int_0^{\infty} e^{-2t} e^{-i\omega t} dt$$

$$g(\omega) = \int_0^{\infty} e^{-2t-i\omega t} dt$$

$$g(\omega) = \int_0^{\infty} e^{-(2+i\omega)t} dt$$

$$g(\omega) = \left[\frac{e^{-(2+i\omega)t}}{-(2+i\omega)} \right]_0^{\infty}$$

$$g(\omega) = \frac{-1}{2+i\omega} [e^{-\infty} - e^{-0}]$$

$$g(\omega) = \frac{-1}{2+i\omega} \left[\frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$g(\omega) = \frac{-1}{2+i\omega} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$g(\omega) = \frac{-1}{2+i\omega} [0 - 1]$$

$$g(\omega) = \frac{1}{2+i\omega}$$

$$|g(\omega)| = \frac{1}{\sqrt{2^2 + \omega^2}}$$

$$|g(\omega)| = \frac{1}{\sqrt{4 + \omega^2}} \quad \text{Answer}$$

Example 44: Find Fourier Transform of

$$f(t) = 1 + \frac{t}{a} \quad ; -a < t < 0$$

$$= 1 - \frac{t}{a} \quad ; 0 < t < a$$

$$= 0 \quad ; \text{otherwise}$$

Answer:

We have,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [g(\omega)] e^{i\omega t} d\omega$$

$$\text{Where } g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\begin{aligned}
g(\omega) &= \int_{-\infty}^{-a} f(t)e^{-i\omega t} dt + \int_{-a}^0 f(t)e^{-i\omega t} dt + \int_0^a f(t)e^{-i\omega t} dt + \int_a^{\infty} f(t)e^{-i\omega t} dt \\
g(\omega) &= \int_{-\infty}^{-a} 0 \cdot e^{-i\omega t} dt + \int_{-a}^0 \left(1 + \frac{t}{a}\right) \cdot e^{-i\omega t} dt + \int_0^a \left(1 - \frac{t}{a}\right) \cdot e^{-i\omega t} dt + \int_a^{\infty} 0 \cdot e^{-i\omega t} dt \\
g(\omega) &= \int_{-a}^0 \left(1 + \frac{t}{a}\right) \cdot e^{-i\omega t} dt + \int_0^a \left(1 - \frac{t}{a}\right) \cdot e^{-i\omega t} dt \\
g(\omega) &= \int_{-a}^0 1 \cdot e^{-i\omega t} dt + \int_{-a}^0 \frac{t}{a} \cdot e^{-i\omega t} dt + \int_0^a 1 \cdot e^{-i\omega t} dt + \int_0^a \left(-\frac{t}{a}\right) \cdot e^{-i\omega t} dt \\
g(\omega) &= \int_{-a}^0 e^{-i\omega t} dt + \frac{1}{a} \int_{-a}^0 t \cdot e^{-i\omega t} dt + \int_0^a e^{-i\omega t} dt - \frac{1}{a} \int_0^a t \cdot e^{-i\omega t} dt \\
g(\omega) &= \int_{-a}^0 e^{-i\omega t} dt + \int_0^a e^{-i\omega t} dt + \frac{1}{a} \int_{-a}^0 t \cdot e^{-i\omega t} dt - \frac{1}{a} \int_0^a t \cdot e^{-i\omega t} dt \dots \dots \dots (i)
\end{aligned}$$

Now,

$$\begin{aligned}
&\int t \cdot e^{-i\omega t} dt \\
&= t \int e^{-i\omega t} dt - \int \left\{ \frac{d}{dt}(t) \int e^{-i\omega t} dt \right\} dt \quad \left[\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx \right] \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] - \int \left\{ 1 \cdot \left[\frac{e^{-i\omega t}}{-i\omega} \right] \right\} dt \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] + \int \frac{e^{-i\omega t}}{i\omega} dt \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{i\omega} \int e^{-i\omega t} dt \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{i\omega} \left[\frac{e^{-i\omega t}}{-i\omega} \right] \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] - \frac{1}{i^2 \omega^2} e^{-i\omega t} \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] - \frac{1}{-\omega^2} e^{-i\omega t} \quad [i^2 = -1] \\
&= t \left[\frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{\omega^2} e^{-i\omega t} \\
&= \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \quad \dots \dots \dots (ii)
\end{aligned}$$

Putting the result of $\int t \cdot e^{-i\omega t} dt$ in (i), we get

[illegible]

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a} \left[\frac{1}{\omega^2} - \frac{1}{i \omega} a e^{i \omega a} - \frac{1}{\omega^2} e^{i \omega a} \right] - \frac{1}{a} \left[\frac{1}{-i \omega} t e^{-i \omega t} + \frac{1}{\omega^2} e^{-i \omega t} \right]_0^a$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a} \left[\frac{1}{\omega^2} - \frac{1}{i \omega} a e^{i \omega a} - \frac{1}{\omega^2} e^{i \omega a} \right] - \frac{1}{a} \left[\frac{1}{-i \omega} a e^{-i \omega a} + \frac{1}{\omega^2} e^{-i \omega a} - \frac{1}{-i \omega} . 0 . e^{-i \omega . 0} - \frac{1}{\omega^2} e^{-i \omega . 0} \right]$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a} \left[\frac{1}{\omega^2} - \frac{1}{i \omega} a e^{i \omega a} - \frac{1}{\omega^2} e^{i \omega a} \right] - \frac{1}{a} \left[\frac{1}{-i \omega} a e^{-i \omega a} + \frac{1}{\omega^2} e^{-i \omega a} - \frac{1}{\omega^2} e^{-0} \right]$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a} \left[\frac{1}{\omega^2} - \frac{1}{i \omega} a e^{i \omega a} - \frac{1}{\omega^2} e^{i \omega a} \right] - \frac{1}{a} \left[\frac{1}{-i \omega} a e^{-i \omega a} + \frac{1}{\omega^2} e^{-i \omega a} - \frac{1}{\omega^2} . 1 \right]$$

$$[e^{-0} = \frac{1}{e^0} = \frac{1}{1} = 1]$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a} \left[\frac{1}{\omega^2} - \frac{1}{i \omega} a e^{i \omega a} - \frac{1}{\omega^2} e^{i \omega a} \right] - \frac{1}{a} \left[\frac{1}{-i \omega} a e^{-i \omega a} + \frac{1}{\omega^2} e^{-i \omega a} - \frac{1}{\omega^2} \right]$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \left[\frac{1}{a \omega^2} - \frac{a e^{i \omega a}}{a i \omega} - \frac{e^{i \omega a}}{a \omega^2} \right] - \left[\frac{-a e^{-i \omega a}}{a i \omega} + \frac{e^{-i \omega a}}{a \omega^2} - \frac{1}{a \omega^2} \right]$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega . a}}{i \omega} + \frac{1}{a \omega^2} - \frac{a e^{i \omega a}}{a i \omega} - \frac{e^{i \omega a}}{a \omega^2} + \frac{a e^{-i \omega a}}{a i \omega} - \frac{e^{-i \omega a}}{a \omega^2} + \frac{1}{a \omega^2}$$

$$g(\omega) = \frac{1}{i \omega} (e^{+a i \omega} - e^{-i \omega a}) + \frac{1}{a \omega^2} + \frac{1}{a \omega^2} + \frac{a e^{-i \omega a}}{a i \omega} - \frac{a e^{i \omega a}}{a i \omega} - \frac{e^{i \omega a}}{a \omega^2} - \frac{e^{-i \omega a}}{a \omega^2}$$

$$g(\omega) = \frac{1}{i \omega} (e^{+a i \omega} - e^{-i \omega a}) + \frac{2}{a \omega^2} + \frac{a e^{-i \omega a}}{a i \omega} - \frac{a e^{i \omega a}}{a i \omega} - \frac{e^{i \omega a}}{a \omega^2} - \frac{e^{-i \omega a}}{a \omega^2}$$

$$g(\omega) = \frac{1}{i \omega} (e^{+a i \omega} - e^{-i \omega a}) + \frac{2}{a \omega^2} + \frac{e^{-i \omega a}}{i \omega} - \frac{e^{i \omega a}}{i \omega} - \frac{e^{i \omega a}}{a \omega^2} - \frac{e^{-i \omega a}}{a \omega^2}$$

$$g(\omega) = \frac{e^{+a i \omega}}{i \omega} - \frac{e^{-i \omega a}}{i \omega} + \frac{2}{a \omega^2} + \frac{e^{-i \omega a}}{i \omega} - \frac{e^{i \omega a}}{i \omega} - \frac{e^{i \omega a}}{a \omega^2} - \frac{e^{-i \omega a}}{a \omega^2}$$

$$g(\omega) = \frac{2}{a \omega^2} - \frac{e^{i \omega a}}{a \omega^2} - \frac{e^{-i \omega a}}{a \omega^2}$$

$$g(\omega) = \frac{1}{a \omega^2} (2 - e^{i \omega a} - e^{-i \omega a})$$

Answer

Unit step function:

Example 84: $-2u(t-1)$

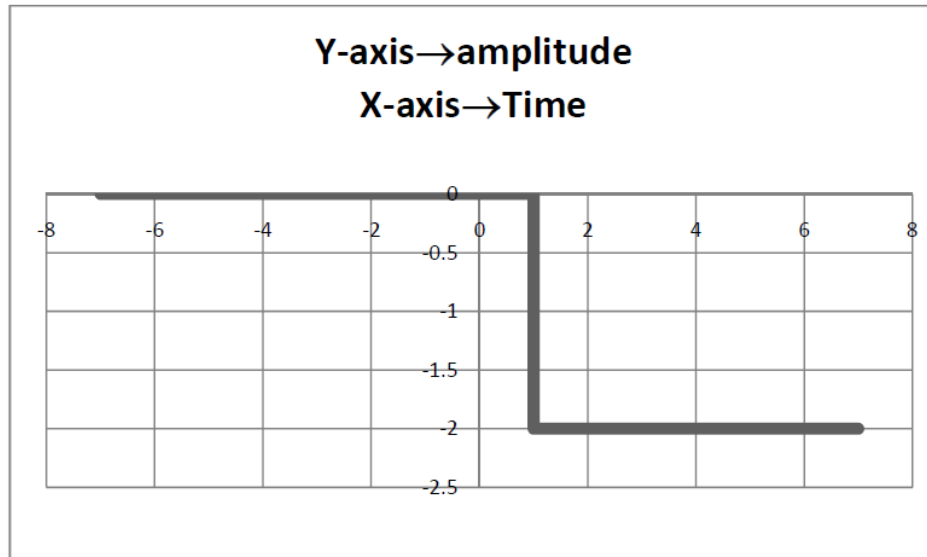
Amplitude

start from

$$y = -2u(t-1) = -2; t \geq 1$$

$$t-1=0 \text{ then } t=1$$

$$=0; t \leq 1$$



Example 89: Given that, $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

Answer:

01. $-u(t+3) \Rightarrow$

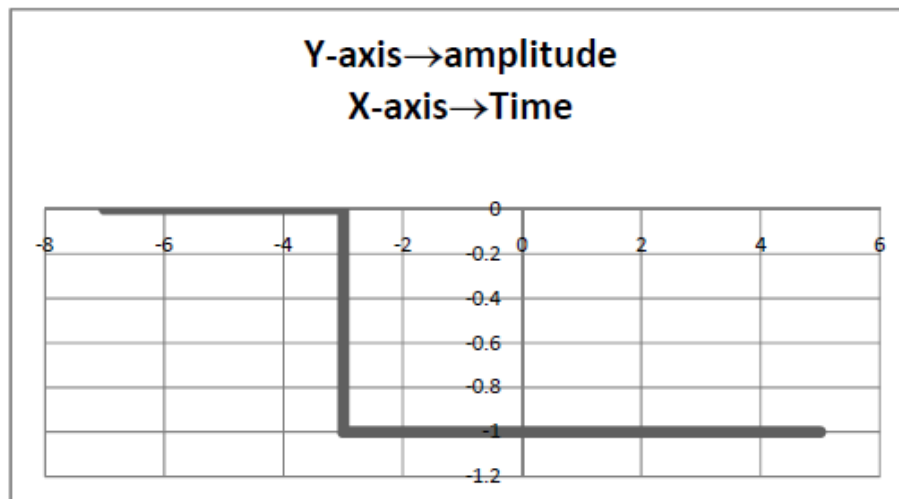
So,

$$-u(t+3) = -1; t \geq -3;$$

$$\text{here, } t+3 = 0$$

$$= 0; t < -3$$

$$\therefore t = -3$$



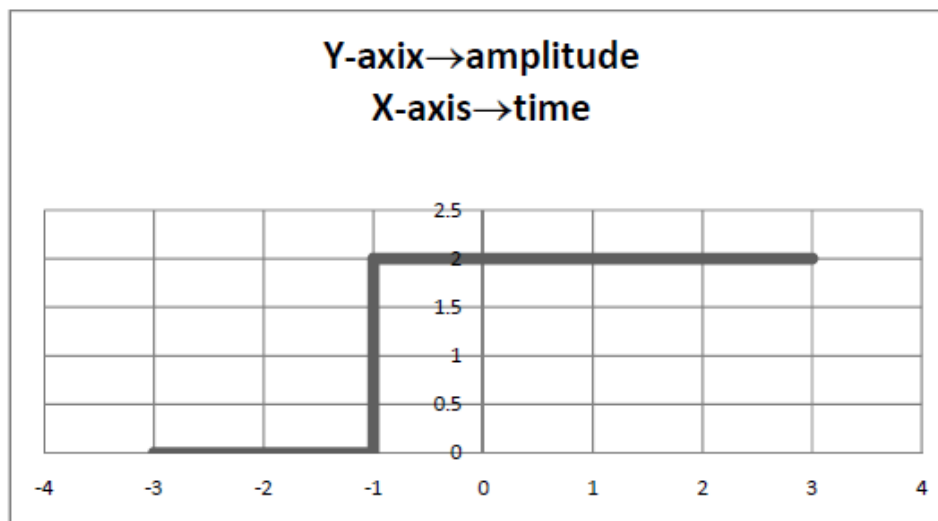
02. $2u(t + 1)$

$$2u(t + 1) = 2; t \geq -1$$

$$\text{here, } t + 1 = 0$$

$$= 0; t < -1$$

$$t = -1$$



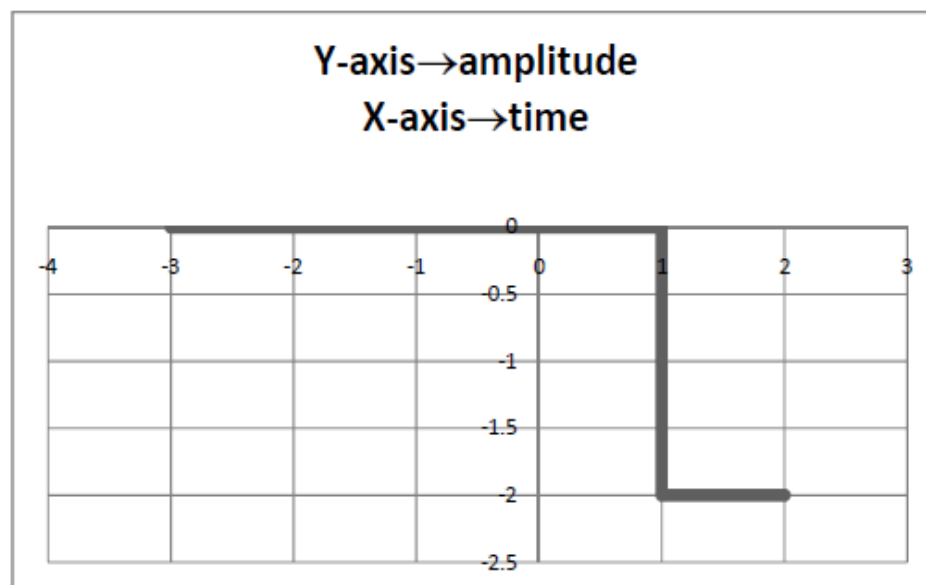
03. $-2u(t - 1)$

$$\therefore -2u(t - 1) = -2; t \geq 1$$

$$\text{here, } t - 1 = 0$$

$$= 0; t < 1$$

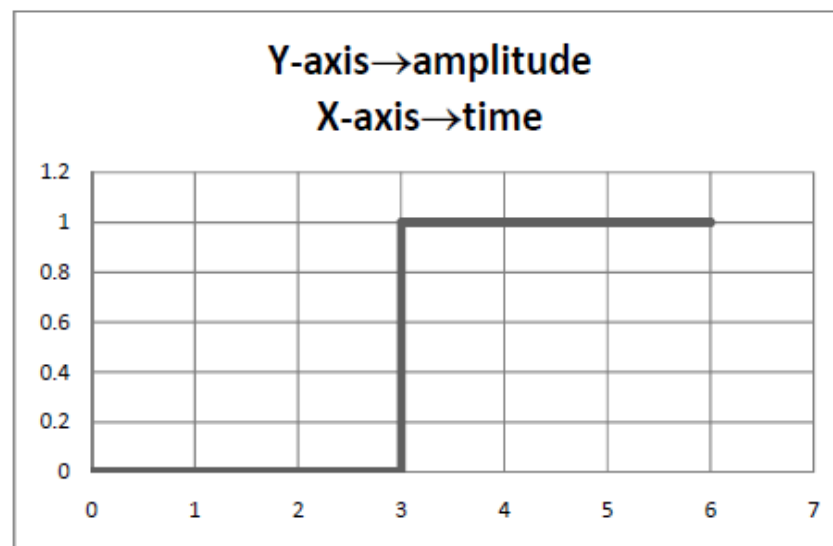
$$t = 1$$



04. $u(t-3) \Rightarrow$

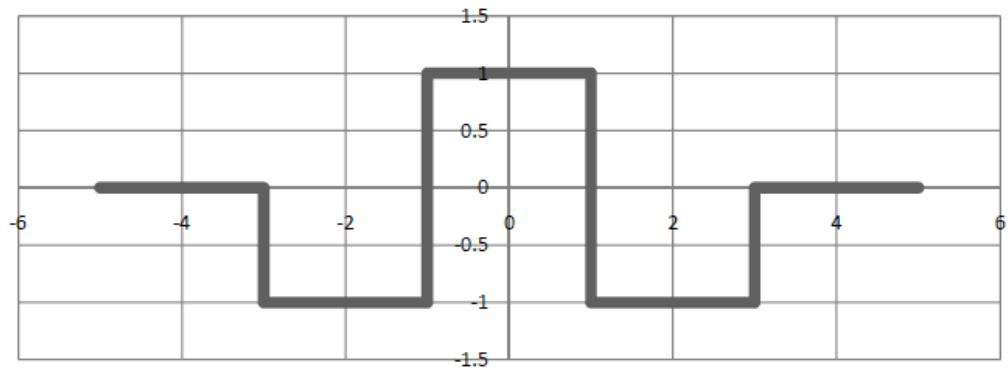
here, $t-3=0$

$t=3$



$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

y-axis=Amplitude $x(t)$
X-Time



Example 91: Find $L[u(t-a)] = \frac{e^{-as}}{s}$

Proof: We have

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt$$

Here, $f(t) = u(t-a)$

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt$$

$$L(u(t-a)) = \int_0^{\infty} u(t-a) e^{-st} dt$$

$$L(u(t-a)) = \int_0^a 0 \cdot e^{-st} dt + \int_a^{\infty} 1 \cdot e^{-st} dt$$

[The unit function $u(t-a)$ is defined as

$$\begin{aligned} u(t) &= 1 & t &\geq a \\ &= 0 & t < a \end{aligned}$$

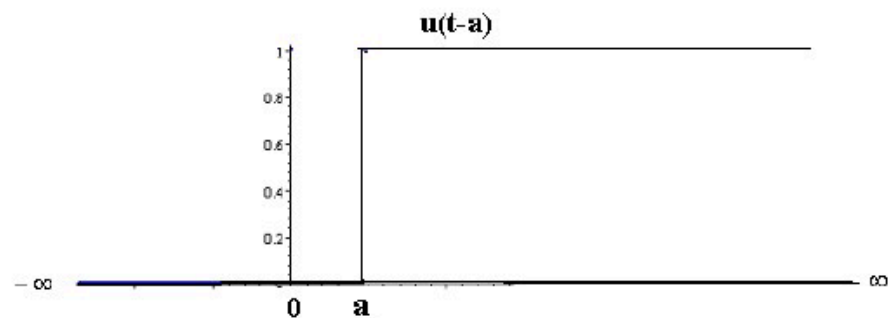


Figure 120

$$\begin{aligned} L(u(t-a)) &= \int_0^a 0 \cdot e^{-st} dt + \int_a^{\infty} 1 \cdot e^{-st} dt \\ &= 0 + \int_a^{\infty} 1 e^{-st} dt \\ &= \int_a^{\infty} e^{-st} dt \\ &= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} \\ &= -\frac{1}{s} \left[e^{-s \times \infty} - e^{-s \times a} \right] \\ &= -\frac{1}{s} \left[e^{-\infty} - e^{-as} \right] \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{s} \left[\frac{1}{e^\infty} - e^{-as} \right] \\
&= -\frac{1}{s} \left[\frac{1}{\infty} - e^{-as} \right] \\
&= -\frac{1}{s} [0 - e^{-as}] \\
&= +\frac{1}{s} [e^{-as}] \\
&= \frac{e^{-as}}{s}
\end{aligned}$$

$$\therefore L(u(t-a)) = \frac{e^{-as}}{s} \text{ (Proved)}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s}$$

Example 92: Express the following function in terms of unit step functions and find its Laplace transform:

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

Solution:

We have

$$L(u(t-a)) = \frac{e^{-as}}{s}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s} \text{------(i)}$$

Given,

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$f(t) = \begin{cases} 8+0; & t < 2 \\ 8-2; & t > 2 \end{cases}$$

$$f(t) = 8 + \begin{cases} 0; & t < 2 \\ -2; & t > 2 \end{cases}$$

$$f(t) = 8 + (-2) \begin{cases} 0; & t < 2 \\ 1; & t > 2 \end{cases}$$

$$f(t) = 8 + (-2) \begin{cases} 1; & t > 2 \\ 0; & t < 2 \end{cases}$$

$$f(t) = 8 + (-2)u(t - 2)$$

$$f(t) = 8 - 2u(t - 2)$$

$$L\{f(t)\} = L\{8 - 2u(t - 2)\}$$

$$L\{f(t)\} = L\{8\} - 2L\{u(t - 2)\}$$

$$L\{f(t)\} = 8L\{1\} - 2L\{u(t - 2)\}$$

$$L\{f(t)\} = 8 \times \frac{1}{s} - 2 \times \frac{e^{-2s}}{s}$$

$$[\because L(1) = \frac{1}{s} \text{ (from example 55)}] \text{ \& } L[u(t - 2)] = \frac{e^{-2s}}{s} \text{ (from example 91)}$$

Problem 32:

Product of $u(t)$ vs. Shifting the Function Along the t -axis

1. $f(t).u(t)$

The $f(t)$ part begins at $t = 0$

2. $f(t).u(t - a)$

The $f(t)$ part begins at $t = a$

3. $f(t - a).u(t)$

The $f(t)$ part has been shifted to the right by a units and begins at $t = 0$

4. $f(t - a).u(t - a)$

The $f(t)$ part has been shifted to the right by a units and begins at $t = a$

Example 93: $g(t) = \sin t . u(t)$

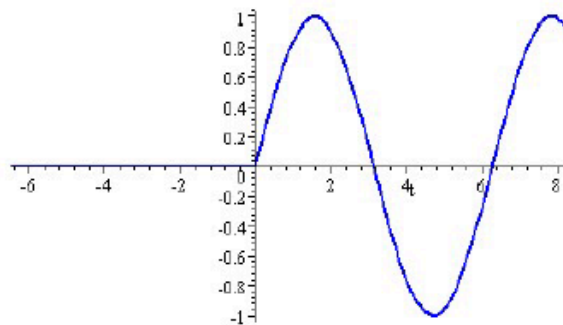


Figure 121: the $\sin t$ part starts at $t = 0$

Ramp function: Ex-90

Example 90:

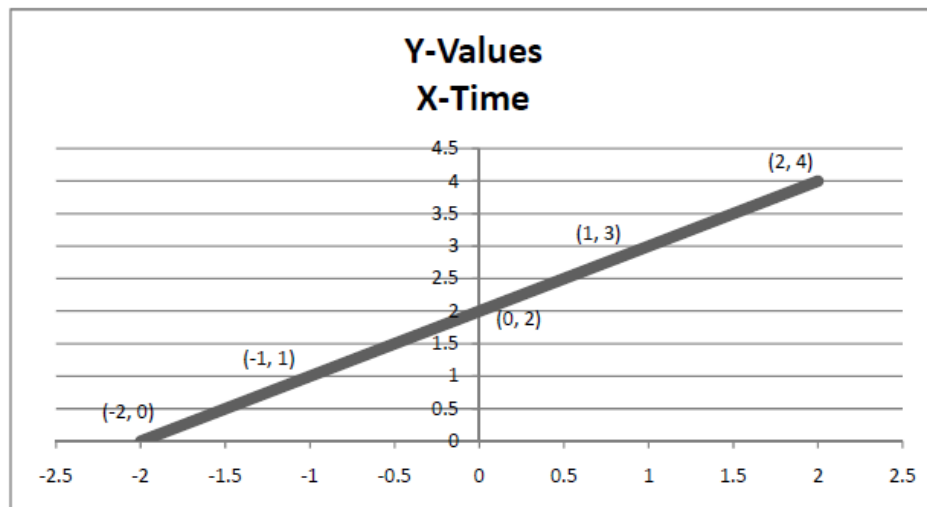
$$x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$$

Solve:

$$\begin{aligned} r(t+2) &= t+2; t \geq -2 \\ &= 0; t < -2 \end{aligned}$$

$$\begin{aligned} \text{Here, } t+2 &= 0 \\ \therefore t &= -2 \end{aligned}$$

t	-2	-1	0	1	2	3
$r(t+2) = t+2$	0	1	2	3	4	5

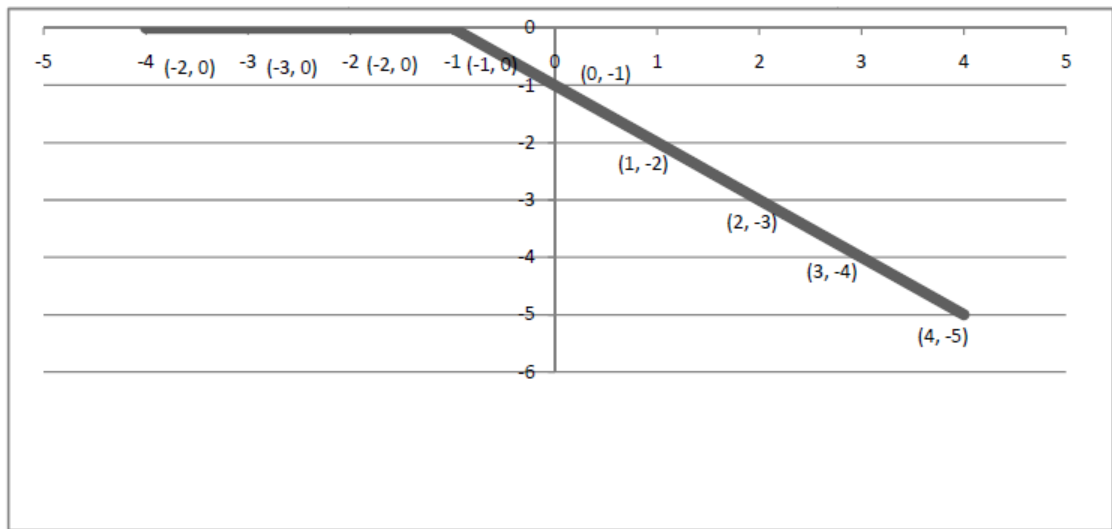


Again,

$$\begin{aligned} -r(t+1) &= -(t+1); t \geq -1 \\ &= 0; t < -1 \end{aligned}$$

$$\begin{aligned} \text{Here, } t+1 &= 0 \\ \therefore t &= -1 \end{aligned}$$

t	-1	0	1	2	3	4
$-r(t+1) = -(t+1)$	0	-1	-2	-3	-4	-5

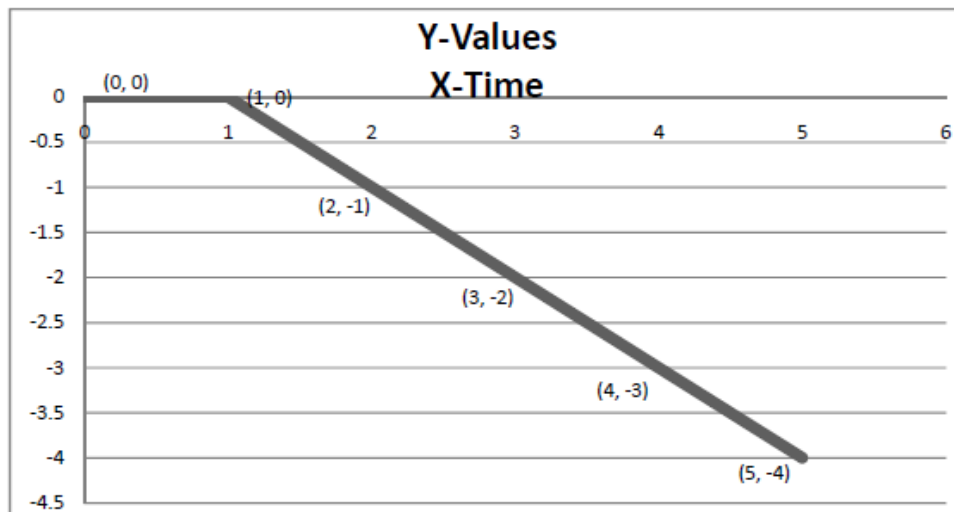


$$\begin{aligned} \therefore -r(t-1) &= -(t-1) && ; t \geq 1 \\ &= 0 && ; t < 1 \end{aligned}$$

$$\text{Here, } t-1 = 0$$

$$\therefore t = 1$$

t	1	2	3	4	5
$-r(t-1) = -(t-1)$	0	-1	-2	-3	-4

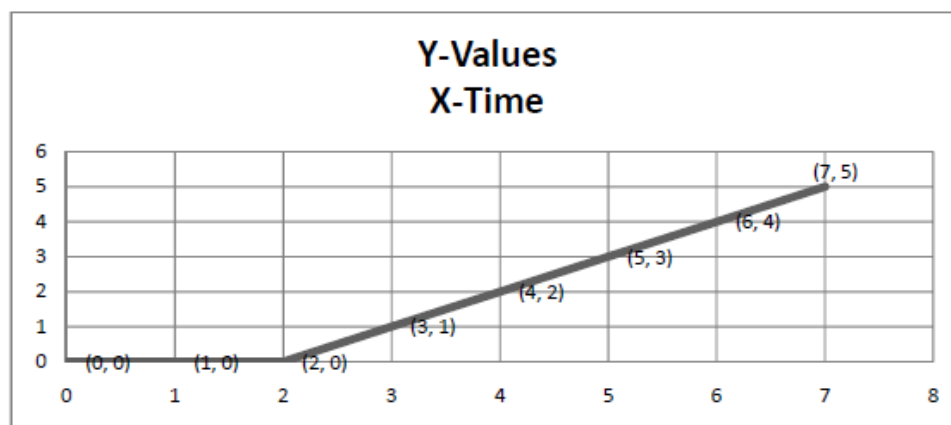


$$r(t-2) = (t-2) \quad ; t \geq 2$$

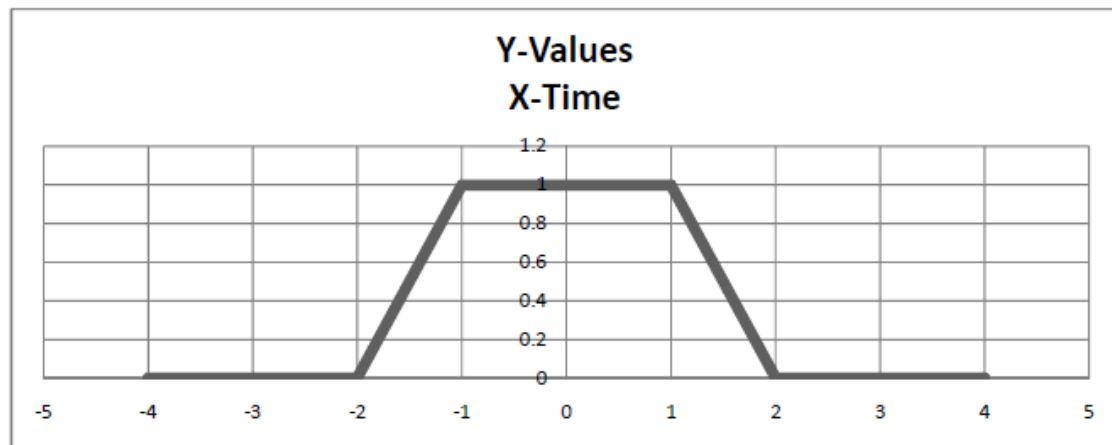
$$= 0 \quad ; t < 2$$

Here, $t-2 = 0$
 $\therefore t = 2$

t	2	3	4	5	6	7
$r(t-2) = t-2$	0	1	2	3	4	5



$$\therefore x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$$



Problem 21: Impulse Function

Impulse Function

The Impulse function $\delta(t)$ is defined as

$$\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{Otherwise} \end{cases}$$

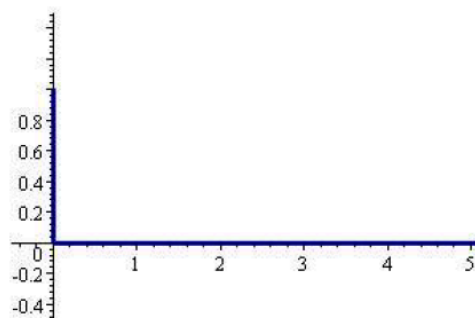
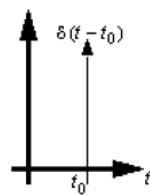


Figure 118



Delta (Impulse) Function

Initial Value Problem (IVP) by Laplace Transform

Example 104: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' + 4Y = 12t \quad Y(0) = 0 \quad Y'(0) = 7$$

Solution

Given,

$$Y'' + 4Y = 12t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' + 4Y = 12t$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

We have,

$$L(f''(t)) = s^2 L\{f(t)\} - s f(0) - f'(0)$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

$$s^2 L\{Y\} - s f(0) - f'(0) + 4L\{Y\} = 12L\{t\}$$

$$s^2y - sf(0) - f'(0) + 4y = 12L\{t\}$$

$$[\text{let}, L\{Y\} = y]$$

$$s^2y - sf(0) - f'(0) + 4y = 12\frac{1}{s^2}$$

$$[\text{let}, L\{t\} = \frac{1}{s^2}]$$

$$s^2y - s.0 - 7 + 4y = 12\frac{1}{s^2}$$

$$[\text{Given}, Y(0) = 0 \quad Y'(0) = 7]$$

$$s^2y - 7 + 4y = \frac{12}{s^2}$$

$$s^2y + 4y - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) = 7 + \frac{12}{s^2}$$

$$y = \frac{7}{(s^2 + 4)} + \frac{12}{s^2(s^2 + 4)}$$

$$y = \frac{7}{(s^2 + 4)} + 3 \times \frac{4}{s^2(s^2 + 4)}$$

$$y = \frac{7}{(s^2 + 4)} + 3 \times \left[\frac{1}{s^2} - \frac{1}{(s^2 + 4)} \right]$$

$$y = \frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore L\{Y\} = y = \frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = L^{-1} \left[\frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right] \right]$$

$$\therefore Y = L^{-1}(y) = L^{-1} \left[\frac{7}{(s^2 + 4)} \right] + L^{-1} \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} - \frac{1}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right] - 3L^{-1} \left[\frac{1}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 4L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$\therefore Y = L^{-1}(y) = 4 \frac{1}{2} L^{-1} \left[\frac{2}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$\therefore Y = L^{-1}(y) = 4 \frac{1}{2} \sin 2t + 3t$$

$$\therefore Y = L^{-1}(y) = 2 \sin 2t + 3t$$

Example 105: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' - 3Y' + 2Y = 4e^{2t} \quad Y(0) = -3 \quad Y'(0) = 5$$

Solution

$$Y = f(t)$$

Given,

$$Y'' - 3Y' + 2Y = 4e^{2t}$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' - 3Y' + 2Y = 4e^{2t}$$

$$L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$$

We have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$$

And

$$\therefore L\{f'(t)\} = sL\{f(t)\} - f(0)$$

$$\therefore L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$$

$$s^2 L\{f(t)\} - s f(0) - f'(0) - 3[sL\{f(t)\} - f(0)] + 2y = 4 \frac{1}{s-2} \quad [\text{let, } L\{Y\} = y \text{ \& } L(e^{at}) = \frac{1}{s-a}]$$

$$s^2 L\{Y\} - s f(0) - f'(0) - 3[sL\{Y\} - f(0)] + 2y = 4 \frac{1}{s-2} \quad [Y = f(t)]$$

$$s^2 y - s f(0) - f'(0) - 3[sy - f(0)] + 2y = 4 \frac{1}{s-2} \quad [\text{let, } L\{Y\} = y]$$

$$s^2 y - s f(0) - f'(0) - 3sy + 3f(0) + 2y = 4 \frac{1}{s-2}$$

$$s^2 y - s\{-3\} - 5 - 3sy + 3(-3) + 2y = 4 \frac{1}{s-2}$$

$$s^2 y + 3s - 5 - 3sy - 9 + 2y = 4 \frac{1}{s-2}$$

$$s^2 y + 3s - 3sy - 14 + 2y = 4 \frac{1}{s-2}$$

$$s^2 y - 3sy + 2y + 3s - 14 = 4 \frac{1}{s-2}$$

$$s^2 y - 3sy + 2y = -3s + 14 + 4 \frac{1}{s-2}$$

$$y(s^2 - 3s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$y(s^2 - 2s - s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$y\{s(s-2) - 1(s-2)\} = -3s + 14 + 4 \frac{1}{s-2}$$

$$y(s-1)(s-2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$y = -3 \frac{s}{(s-1)(s-2)} + 14 \frac{1}{(s-1)(s-2)} + 4 \frac{1}{(s-2)} \frac{1}{(s-1)(s-2)}$$

$$y = \frac{-3s}{(s-1)(s-2)} + \frac{14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s + 14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2}$$

$$y = \frac{(-3s + 14)(s-2) + 4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s^2 + 6s + 14s - 28 + 4}{(s-1)(s-2)^2}$$

$$y = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2}$$

Applying partial fraction

Let,

$$y = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2} = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-2)^2}$$

.....

.....

$$y = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore \mathcal{L}\{Y\} = y = \frac{-3s^2 + 6s + 14s - 24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2}$$

$$\therefore Y = \mathcal{L}^{-1}(y) = -7 \frac{1}{2} \mathcal{L}^{-1}\left[\frac{1}{(s-1)}\right] + 4 \mathcal{L}^{-1}\left[\frac{1}{(s-2)}\right] + 4 \mathcal{L}^{-1}\left[\frac{4}{(s-2)^2}\right]$$

$$\therefore Y = \mathcal{L}^{-1}(y) = -7e^t + 4e^{2t} + 4te^{2t}$$